

Implementation of Lean Manufacturing Principles to increase Productivity in SMEs in the manufacturing sector of clothing

Marco Arica-Hernandez
Facultad de Ingeniería, Programa de
Ingeniería de Gestión Empresarial,
Universidad Peruana de Ciencias
Aplicadas, Lima, Perú
u201415106@upc.edu.pe

Sebastian Llagas-Llontop
Facultad de Ingeniería, Programa de
Ingeniería de Gestión Empresarial,
Universidad Peruana de Ciencias
Aplicadas, Lima, Perú
u201616146@upc.edu.pe

Juan Carlos Quiroz-Flores
Facultad de Ingeniería, Programa de
Ingeniería de Gestión Empresarial,
Universidad Peruana de Ciencias
Aplicadas, Lima, Perú
pcijqui@upc.edu.pe

ABSTRACT

SMEs in the textile sector face many problems in their production flows, mainly due to the lack of production management systems caused by poor management of the production chain. Therefore, a diagnostic analysis is carried out in a textile SME to evaluate and define the deficiencies and factors that affect its competitiveness, which began with the analysis of the current situation of the company, where it was established the existence of poor-quality management, high waiting times and lack of procedures. Therefore, the use of lean manufacturing tools such as Jidoka, Single Minute Exchange of Die with respect to the production line and process management for the measurement and control of operations in the production area is proposed. The incorporation of these tools in block allows to decrease the rates of defective products, the excess of operative work and the set-up of the machines for the change of model. The main result of the research was that production increased to 0.091 und/PEN.

CCS CONCEPTS

• Applied computing; • Enterprise computing; • Business process management;

KEYWORDS

Lean manufacturing, SMED, Textile Industry, Productivity

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1 INTRODUCTION

In recent years, the demand for the textile industry has had a continuous and sustainable increase, which leads small and medium-sized companies in the field to face new challenges with respect to cost,

time and product quality, all this directly related to the production area; in the same way, obtain an optimal profitability and profit margin [1]. The textile sector in the country contributes around 740,000 jobs and imports worth US \$ 1,400 million, this symbolizes 10% of the Peruvian industry [2]. According to the study carried out by the National Institute of Statistics and Informatics (INEI) in 2018, the most representative activities were the textile and leather sector (30.6%), the food and beverage sector (17.4%), development of metal products (16.2%) and others [3]. The textile industry encompasses several activities ranging from fiber treatment to yarn, fabric, and clothing manufacturing. [4].

As primary products of the Peruvian textile sector are clothing, among which are polo shirts, pants, sweaters, shirts, etc. The main problem of the case study presented, the subject of the research, is the low productivity generated by the deficient quality management, high waiting times and the high variation in production times. The optimization of these internal processes requires tools, methodologies, and procedures for an adequate application of resources. Likewise, the Lean Manufacturing and process management tools will be identified in greater detail in the state of the art. The first of them establishes tools that reduce the high rate of defective products, which are mainly caused by the lack of training in the workforce and the maintenance procedures in their machinery. The last of them determines the standardization of the processes, facilitating the knowledge of how to carry out a certain process, speeding up its time with the least number of resources possible, maximizing savings with respect to production and reducing costs with respect to the reprocessing of the garments. A joint model of Lean Manufacturing tools and process management would help to establish standardized procedures in the production area of any organization, reduce waste, as well as reprocessed products. Finally, production processes must be managed for a better analysis of results and a standardization of processes must be carried out to reduce labor costs, update procedures, optimize spaces, etc.

This research is motivated by the joint application of the tools, to implement a model that focuses on the systematic reduction of defective products and increases the productivity of the area. This model includes labor, materials, machinery, and internal procedures, all of them of vital importance for a correct application of the proposed tools to maximize results in the manufacture of garments. To present the current proposal, this article has been divided into the following sections: State of the art, contribution, validation, and conclusions.

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2 STATE OF THE ART

Small and medium-sized companies that are within the textile sector are in an increasingly competitive context and to face this, a change is necessary that allows them to reduce costs, lighten processes, increase quality, among others. To achieve this, it is necessary to implement methodologies that allow them to innovate and improve their production system, since currently companies must meet customer expectations and deliver a quality product at a good price. The scientific articles registered and subsequently selected in the state of the art were classified by the main problems encountered by various authors with respect to low productivity in the clothing area. Next, the classification of the categories will be presented.

2.1 Critical Process Mapping

Every organization seeks to reduce costs to be more competitive and be able to maintain profitability, they are adopting the lean manufacturing (LM) methodology and its tools to achieve the reduction of waste and consequently its costs. [5] The Value Stream Map (VSM) is an efficient LM tool, helping to identify production stream waste. [6]

The VSM method is widely used in various fields. It is a promising method that is based on the visualization of production processes. However, the method is not perfect, especially for improving production processes. The regulatory method can improve the efficiency of companies. Due to the characteristics of each specific industry and each specific company, there is no method to evaluate efficiency and optimize processes. [7].

2.2 Quality Control in the process manufacture

Quality should not only be used to measure final products, on the contrary, this tool must be present throughout the entire production process. The definition of control in the quality process falls on the amount of waste that is generated in a production process, the higher the quality control, the lower the waste generated. That is why the key to better productivity is through quality control of a process and adapting the production and capacity times of the line. [8, 9]

The garment industry is a labor-intensive industry, which has motivated brands to outsource production to salaried developed countries in Asia, Africa and South America. [10] This change will generate that organizations seek to lower their labor costs even more, so they would have to move to even lower countries, which leads garment manufacturers to adopt a new method of manufacturing to improve product quality in the production process.

2.3 Setup in clothing area

The single minute exchange to die (SMED) tool is included within the lean production philosophy and its objectives are to reduce waste in the production system and standardize machine change times [11]. These changes refer to any planned modification, for example, tool or material change, product change, and a change in the production reference [12]. Within its concept it is established that any change in the machine must take a certain specific time and the tasks must be standardized for that change. The SMED methodology is based on three pillars: study of external factors

for the application, analysis of internal factors, validation of the measurement method.

2.4 Process Standardization

Supported by the concepts and definitions of the standard work methodology that supports the lean philosophy, the standardization of processes supports the reduction or elimination of activities that do not generate added value, help to improve productivity, and are associated with continuous improvement actions. to eliminate waste in any production process. [13] Under this concept, BPM is the discipline of how to guarantee the performance of processes in an organization to produce consistent results and to understand the opportunities to improve performance.

Various authors propose a model for a BPM life cycle that is represented as a continuous cycle that costs several phases: Identification, discovery, analysis, redesign, implementation and monitoring of processes. Process management becomes very important for an organization to function properly and have a positive effect, several studies indicate a positive correlation between BPM and organizational success [14].

The references according to each author mentioned in the state of the art are summarized in Table 3, classified according to the tools that have been implemented in this article.

3 MODEL PROPOSED

At this point, the phases of the model are developed to improve low productivity in the clothing area for SMEs companies. Built-in tools aim for constant productivity improvement over higher-demand products. Table 1 shows a comparative matrix of components of the proposal vs. state of the art.

The main objective of the model proposed in this study is to combine the tools previously shown in the state of the art, to obtain greater productivity in the clothing area, reduce defective garments and reduce processed product. For proper planning, the design of the proposal was divided into 5 stages, considering basic concepts of project management. Figure 1. First, a diagnosis of the case study was carried out using the VSM as the main tool, that is, collecting data from the textile company, analyzing them and being able to know the main problems of low productivity and their direct causes. Second, a review of previous successful cases was carried out as antecedents, in the same way, the possible tools and methodologies that related to the general problem were constantly investigated that help to quantify the economic impact of the causes and elaborate the design of a viable proposition. Third, the application of tools such as Jidoka, SMED and BPM which will help implement standardization and control in the manufacturing processes. Next, process management was applied to control and monitor the improved and implemented processes. Likewise, the proposed indicators will be reviewed to determine the viability of the project and its impact, recording the improvements that were obtained. Finally, a comparative report of the two scenarios will be prepared: current situation vs. improvement situation.

It should be highlighted that the current study is unique in that it incorporates many methods that have independently demonstrated to be effective. The sort of small manufacturing enterprises in which the tools were used, as well as the industry to which they were

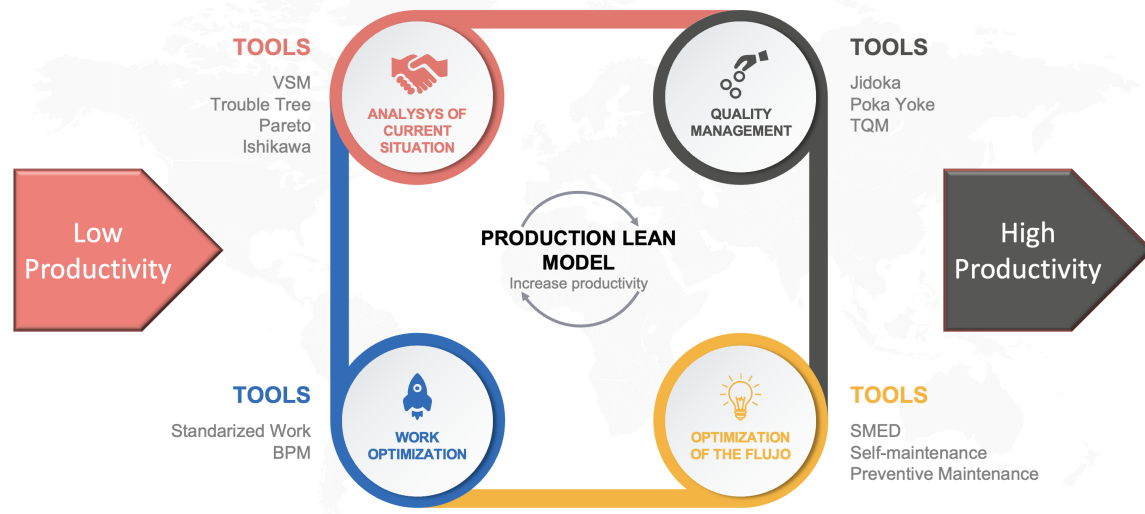


Figure 1: Proposed Improvement Model

Table 1: Indicators Table

Indicator	Formula
% Defectius product	$\frac{\text{Defectious units}}{\text{Total Produced Units}} \times 100$
Productive capacity	$\frac{\text{Available time}}{\text{Cycle time}}$
Preparation time	$\sum(\text{Time for machine preparation activities})$
% Availability	$\frac{\text{net production time}}{\text{Time scheduled to produce}} \times 100$

committed, such as footwear, textiles, and plastic, were studied to conduct an analysis of the problem. Therefore, it was established which instruments should be used to attain the desired outcomes.

3.1 Model Indicators

Each of these indicators is essential to measure the improvement that is being implemented with Lean Manufacturing tools. The main indicator that is measured is the KPI of productivity since the main objective of the company is to increase this indicator to provide better service to customers.

4 VALIDATION

This model was validated by simulating the production process of the case study in Arena Software. The company in question presents problems of delays in the delivery of its orders, due to poor management of its production process, which generates an annual average of 5.74% of defective product, an increase in the production time of the garments and therefore a greater effort of its labor force, which incurs in a higher cost. The implementation of the improvement model was carried out in the case study's apparel area, which manufactures trouser-type garments, whose analysis left it as the company's standard product.

4.1 Initial Diagnostic

The case study shows a technical gap with respect to the company's productivity (0.031 Und/PEN) and the average productivity level in the textile sector (0.13 Und/PEN). Similarly, the average monthly cost per reprocess in the company is 3,904.82 PEN, this monetary amount is represented by about 5.74% of defective products. Likewise, the standard production times for a pair of pants are highly variable, which leads to higher overtime labor costs; in the last year the company spent approximately 19,482 PEN in paid overtime. Lastly, the absence of a procedure plan generates a high number of machine stoppages, with an average changeover time of 4.3 hours, which results in high production waiting times.

4.2 Contrast of validation and initial diagnosis

In the case of the validation of this research, a methodological model and a simulation were developed during three months from March to May. Therefore, the research is divided into phases, in phase one, the VSM was used to know the entire production process, to understand in detail the flow and processes in the production area. In addition, to know the area where the critical process is located. In the second phase, after studying the critical process, the SMED tool was used to reduce the time of one of the critical problems, changeover time, in the production area. As a last phase, BPM was

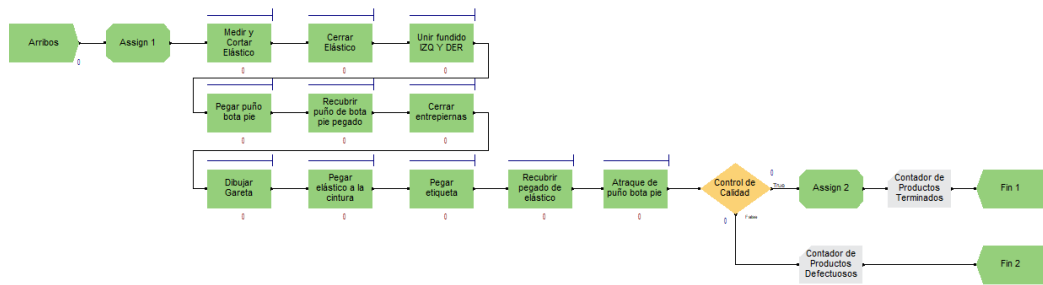


Figure 2: Model Simulation in Arena Software

Table 2: Indicators Table Situation Actual vs Expected

Indicators	Actual	Expected
% Defectius product	5.78%	4%
Productive capacity	194 unt/day	200 und/day
Preparation time	4.3h	3h
% Availability	80%	82%

Table 3: Table of Distribution

Station	Distribution	Unit
Arrive	Expo (2.45)	Minutes
Measure and cut elastic	Norm (0.0985,0.0062)	Minutes
Cover with elastic	Norm (1.01,0.00569)	Minutes
Join molts	Norm (0.545,0.0452)	Minutes
Fist to botapie	Tria (0.72,0.786,0.87)	Minutes
Close elastic	Tria (0.09,0.179,0.22)	Minutes
Paste label	Tria (0.59,0.624,0.87)	Minutes
Cover botapie fist	0.79+0.02*BETA (5.78,9.96)	Minutes
Hit elastic to the waist	Norm (0.0985,0.0062)	Minutes
Draw garet	Norm (1.23,0.0034)	Minutes
Fist trap	0.68 +WEIB (0.00976,855)	Minutes
Close crotch	1.18 +WEIB (0.0205,2.92)	Minutes

used as a tool for the implementation in the standardization of production processes. Table 2

4.3 Simulation of the improvement proposal

To validate the results of the proposal, the Arena software was used, which allows the simulation of the proposed system and to automate and sequence the elaboration of tasks within the production process, since there was no previous study of it. For this, through the tools previously described, it is expected to be able to optimize the production process of the organization's apparel area and reduce the cycle time, the number of defective garments, and increase the company's profitability. Once the times were obtained, the data were entered into the input analyzer tool to determine

the mean and standard deviation corresponding to each activity described in the process. Table 3 shows the results obtained, as well as the unit of time in which they work.

For the present study, a simulation model was carried out to corroborate the impact and efficiency of our proposal, considering the following: the scope of the simulation begins in the manufacturing process and the impact of the tools used can be measured in Figure 2 shows the representation of the system developed in the Arena Simulator software. Also, for the simulation of the process, valid data were used with the information provided by the company. Table 4 shows the simulation results compared to the initial diagnosis of the case study:

5 CONCLUSIONS

Through the application of Lean manufacturing and process management tools, it is possible to increase the productivity of the manufacturing area to 0.091 und/PEN. In addition, it was possible to obtain a 3.13% reduction in defective products, a higher production capacity of 214 und/day, a preparation time of 2.8 hours and 85% availability. Finally, it was possible to standardize the line changeover processes in the machinery, allowing better use of human capital and investments made in the production process.

The proposed model helps to develop a sequence of interrelated sequence of interrelated activities that help to identify all identify all the activities involved in each process. a given process. With the standardization of processes within the organization, it allows the organization allows the construction of a cycle of continuous continuous improvement cycle in which resources can be optimized resources and add relevant improvements according to the type of type of analysis to be carried out. With the SMED tool it is possible to optimize the time to change a machine in line, either by model change or deterioration, this change of model or deterioration, this allows the production line to be more the production line to be more dynamic in production production line to be more dynamic and to produce a higher quantity of garments desired quantity of garments.

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Table 4: Indicators Table Situation Actual vs Simulation

Indicators	Actual	Expected	Simulation
% Defectious product	5.74%	4%	3.13%
Productive capacity	194 unt/day	200 unt/day	214 unt/day
Preparation time	4.3h	3h	2.8h
% Availability	80%	82%	85%

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